

Desk controller

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Change history

1.0.0	23/09/2026	First release
1.1.0	18/10/2026	Migration to KiBot template
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD

System overview

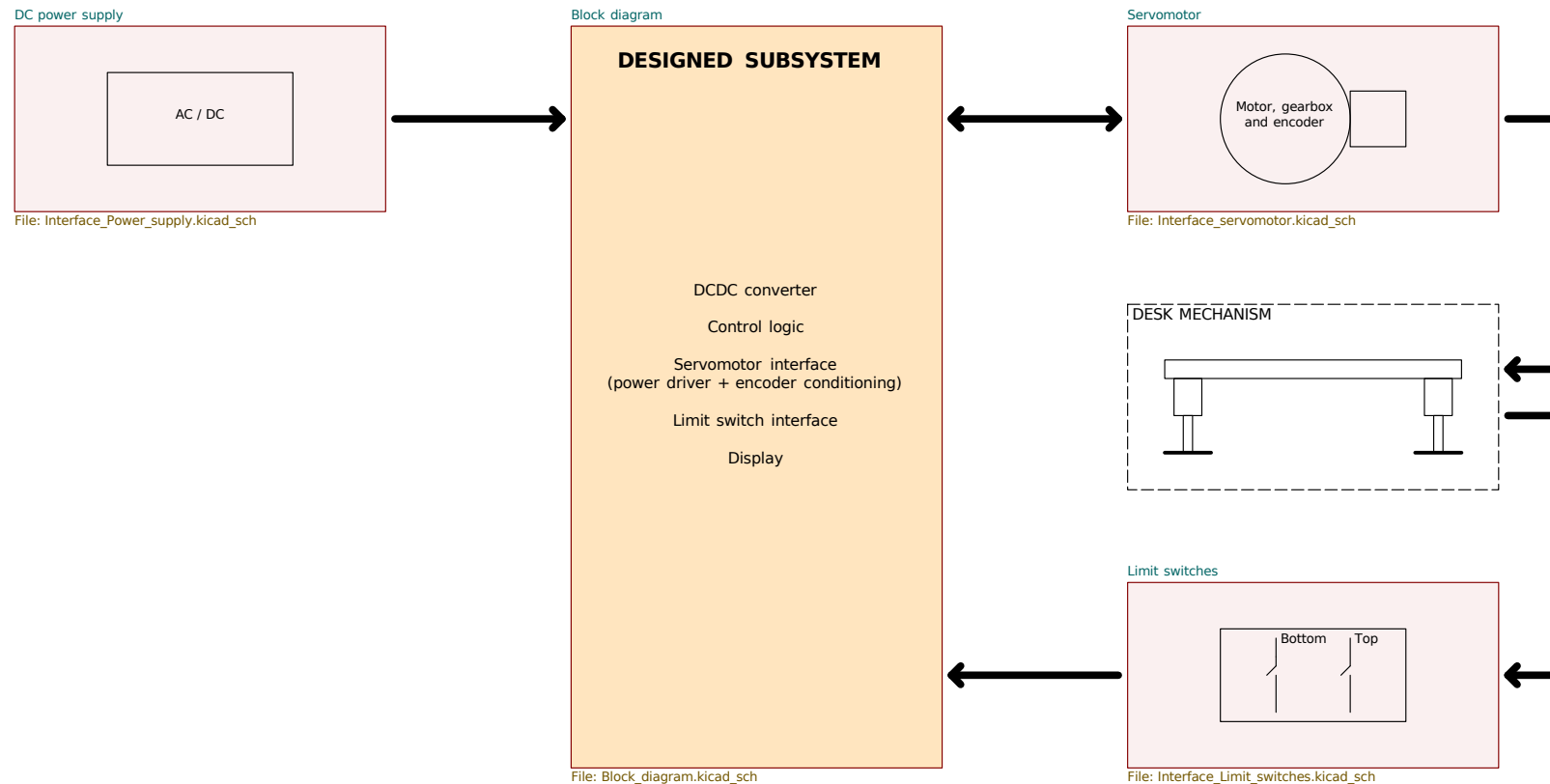


File: System_overview.kicad_sch



Desk controller		Sheet: Root	
Notes		File: ktest.kicad_sch	
		Rev: 1.0.1	Drwn: FLR
		Mat: Concept	Rev'd: FLR
		Date: 2026-10-06	RLs'd: FLR
		Dwg: NA	Size: A4
		Generated with KiCad	Page: 1/11

System overview



This sheet shows the overall system and the interfaces to external components. External components are shown for system context only and are not part of this schematic. Only the "DESIGNED SUBSYSTEM" is covered by this schematic.

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	Desk controller		Sheet: System overview	
	Notes		File: System_overview.kicad_sch	
			Rev: 1.0.1	Drwn: FLR
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DC power supply

Interface specification

Barrel connector (Dell type, with ID pin)

- Pin
- 1 Supply - 19.5 V, 3 A (inner shell)
 - 2 ID (internal pin)
 - 3 GND (outer shell)
 - SH Shield

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 ChurrasCod Electronics	Desk controller		Sheet: DC power supply	
	Notes		File: Interface_Power_supply.kicad_sch	
			Rev: 1.0.1	Drwn: FLR
			Mat: Concept	Rev'd: FLR
			Date: 2026-10-06	RLs'd: FLR
			Dwg: NA	Size: A4
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Interface specification

JST B6B-XH-A / JST XHP-6 connector

Pin	Description
1	Encoder B output (VCC)
2	Encoder A output (VCC)
3	GND
4	VCC (3.3 - 5V)
5	Motor B (3 A, 12-24 V)
6	Motor A (3 A, 12-24 V)

 ChurrasCod Electronics	Desk controller		Sheet: Servomotor	
	Notes		File: Interface_servomotor.kicad_sch	
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			Mat: Concept	Rev'd: FLR
			Date: 2026-10-06	RLs'd: FLR
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Limit switches

Interface specification

JST B4B XH A / JST XHP-4 connector

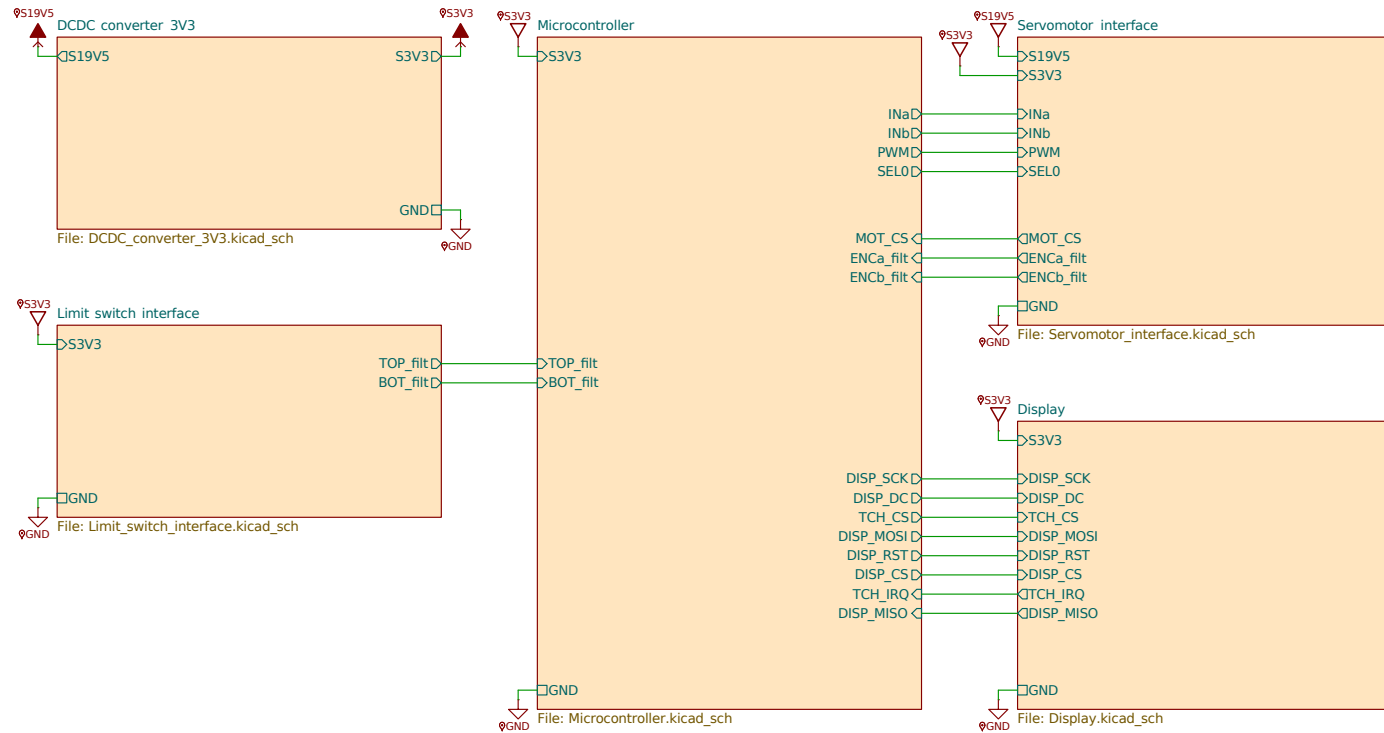
Pin Description

- 1 Bottom limit switch (Normally open, switch closes to GND)
- 2 GND
- 3 GND
- 4 Top limit switch (Normally open, switch closes to GND)

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 ChurnasCod Electronics	Desk controller		Sheet: Limit switches	
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			Mat: Concept	Rev'd: FLR
			Date: 2026-10-06	RLs'd: FLR
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Block diagram



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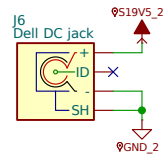
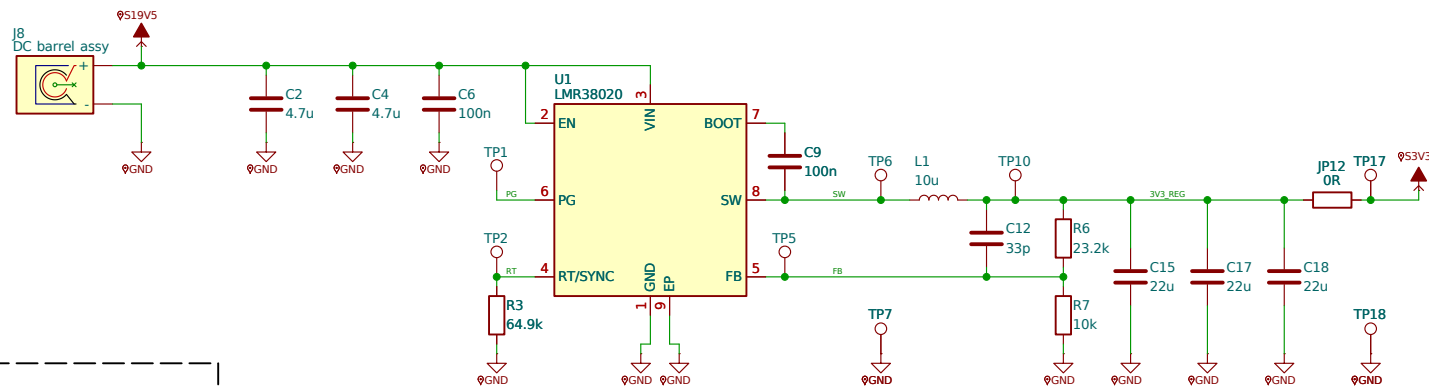
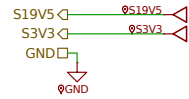
Desk controller

Notes

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DCDC converter 3V3

Block interface



J6 is fitted to a small "standalone" PCB which will then be soldered into the "main" PCB. This standalone PCB + connector are what make the "DC barrel assy" used in J8 position.



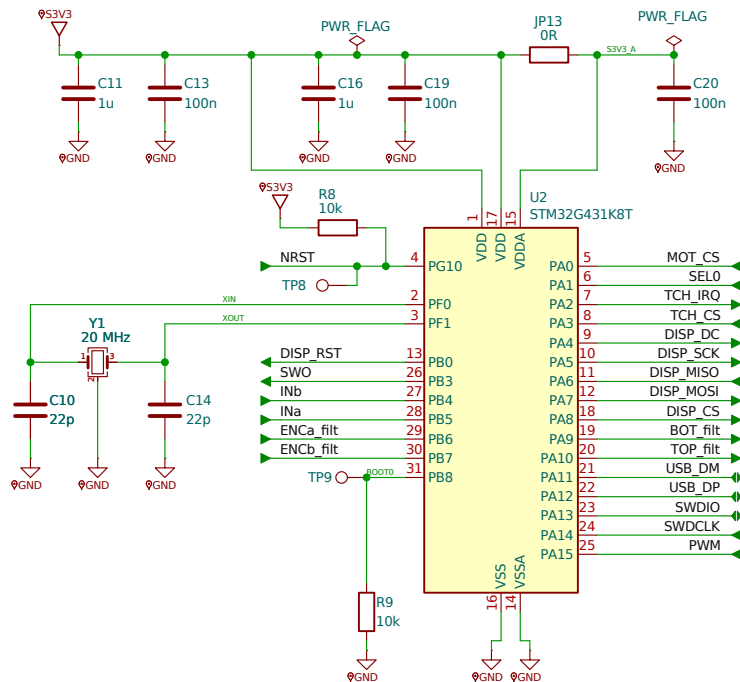
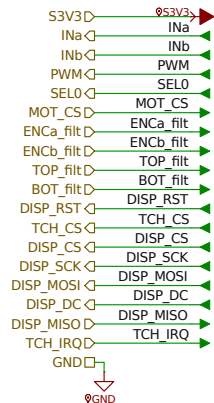
Desk controller

Notes

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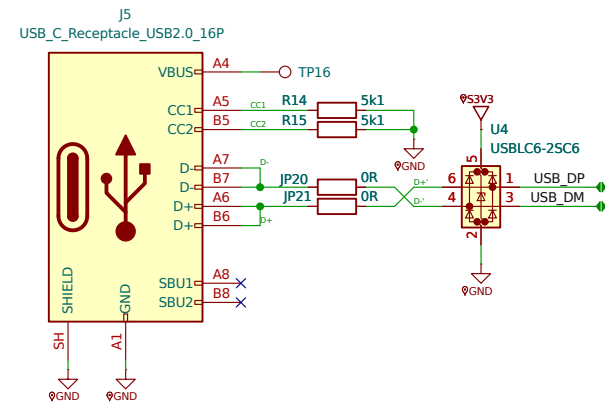
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Block interface

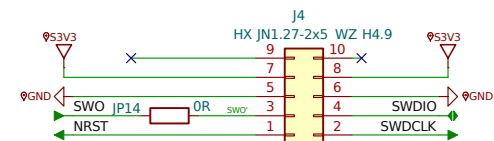


Microcontroller

USB interface (serial comms port)



SWD header



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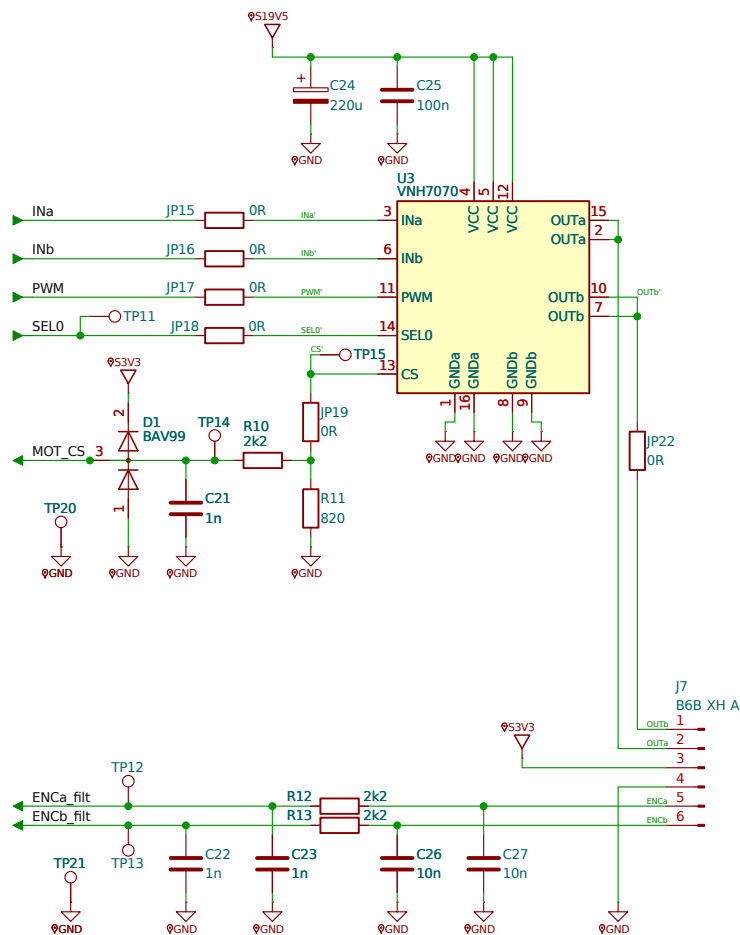
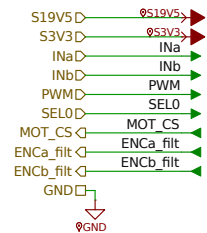
Desk controller

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Servomotor interface

Block interface



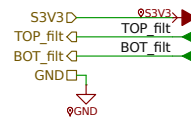
Desk controller

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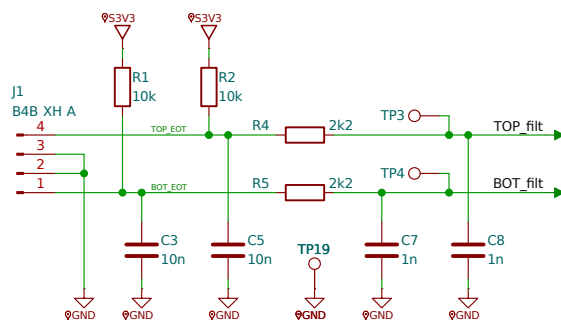
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Block interface



Limit switch interface



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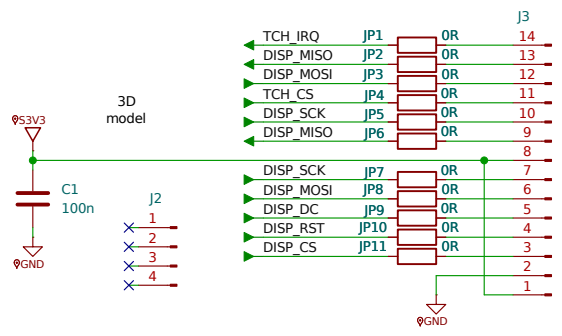
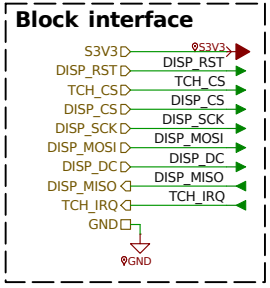


Desk controller

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Display



TFT1
TFT_2.4in_320x240

The display will not be directly soldered to the PCB. Instead, generic 2.54 mm pitch sockets J2 and J3 will be used as connectors.

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